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AnyTime-CD: Self-supervised change detection in arbitrary-length dense time series of high-resolution remote sensing images

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ABSTRACT

With the continuous enhancement of the observation capabilities of remote sensing satellites, dense multi-temporal high-resolution remote sensing images have become an important data source for monitoring surface changes. However, multi-temporal change detection (MTCD) remains challenging due to inconsistent time intervals, spatial misalignments, and a scarcity of labeled data, all of which hinder the generalization and practical deployment of deep learning models in large-scale applications. To address the above challenges, this paper proposed a multi-task self-supervised change detection framework (i.e., AnyTime-CD), which supports the input of time series of any length and can jointly model temporal-spectral-spatial features without manual annotation. This framework consists of three sub-tasks: temporal dynamic modeling, spectral semantic consistency learning, and spatial feature alignment, which respectively address the issues of irregular time interval, spectral perturbation, and spatial offset, thereby collaboratively optimize network performance. We evaluated AnyTime-CD on two globally distributed multi-temporal high-resolution datasets. The experimental results showed that, under an unsupervised setting, this method achieved performance close to or even surpassing that of the supervised methods. Compared with the best-performing self-supervised method, it achieved relative gains of over 13% (spatial) and 8% (temporal) in F1 scores (using a one-frame tolerance strategy), respectively. After sample fine-tuning, it further achieved F1 scores of 86.40% (spatial) and 79.50% (temporal), exceeding the state-of-the-art supervised method by 4.44% and 9.09%, respectively. Furthermore, AnyTime-CD showed good adaptability to different pre-training configurations. It is worth noting that it exhibited strong robustness under medium and low cloud cover conditions and could even utilize cloud perturbations as data augmentation to further enhance performance. In conclusion, AnyTime-CD offers a flexible and label-free solution for MTCD tasks, suitable for complex, dynamic, and interference-prone remote sensing scenarios, and is expected to promote the application of self-supervised methods in surface monitoring.

1. Introduction

Surface monitoring is fundamental to understanding ecological changes and formulating sustainable development strategies (Jones et al., 2011; Huang et al., 2020). Thanks to the advantages of remote sensing in spectral, spatial, and temporal resolution, change detection has become a key task, widely applied in land management (Saha et al., 2024), environmental monitoring (Zhang et al., 2021), disaster assessment (Zhang et al., 2021), and urban planning (Huang et al., 2017).

In recent years, deep learning has become the mainstream approach for remote sensing change detection due to its ability to automatically extract features and model complex spatial-temporal dependencies (Li et al., 2022). Most existing studies focus on bi-temporal change detection (BTCD), which identifies changes by comparing two remote sensing images (Wang et al., 2024; Lv et al., 2024). Although BTCD is computationally efficient and requires less storage, it is difficult to capture the complete dynamics of progressive, periodic or nonlinear changes. With the continuous observation of satellites, dense time series images (TSI) are becoming increasingly abundant, and the research focus is shifting to multi-temporal change detection (MTCD) (He et al., 2024; Hafner et al., 2025). By leveraging temporal continuity, MTCD locates changes but also restores their evolution trajectories (Zhang et al., 2021; Yang et al.,

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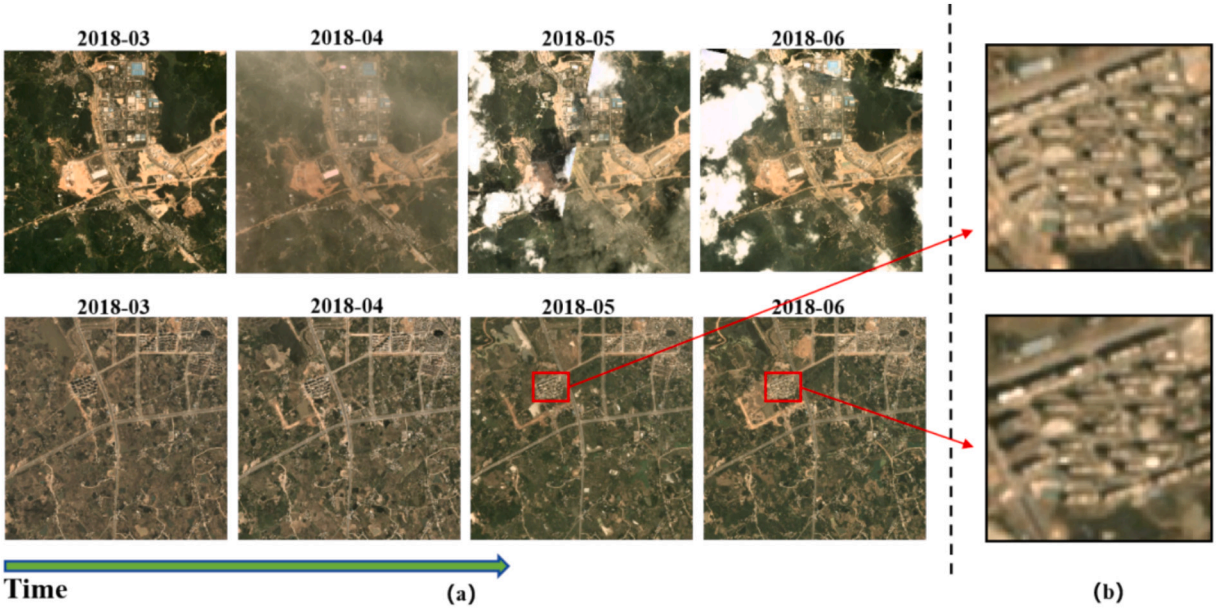


Fig. 1. Examples from two regions (each row represents a region) in the MUDS dataset (Zhang et al., 2021) from March to June 2018. The first row in (a) indicates the number of effective time series for different pixels in the same area is irregular due to cloud interference. (b) is an enlarged view of the local patch in the second row in (a). High-resolution images are difficult to achieve pixel-wise registration, and matching error will be further magnified in multiple time series.

2022; Li et al., 2023), thereby supporting complex applications such as vegetation phenological monitoring, urban expansion assessment, land degradation tracking and ecological disturbance detection.

Although the MTCD method has significant advantages in revealing the dynamic evolution of the surface, it still faces many challenges when applied on a large scale across regions. Firstly, existing methods usually assume that the intervals and lengths of image sequences are consistent, while actual remote sensing time series are often highly irregular, making it difficult to accurately model the time features. In large-scale MTCD, the main causes of these challenges are the temporal irregularities induced by cloud cover, orbit shifts, and on-demand tasking: ① Cloud cover varies by region and season (Zhao et al., 2020), often causing substantial image loss and extended observation gaps (as shown in Fig. 1a); ② The characteristics of satellite orbits cause sampling frequency to be too dense in high-latitude or some trajectory revisit areas and too sparse in other areas (Wulder et al., 2019) ; ③ Commercial satellites are scheduled on demand, with intensive observations of economic hotspots and sparse observations in remote areas. Therefore, the irregular sequence has become the core challenge restricting large-scale and cross-regional deployment of MTCD.

Meanwhile, the emergence of small satellite constellations (e.g., PlanetScope) has significantly improved the spatial and temporal resolution of remote sensing imagery, enabling the detection of fine-scale changes such as new building construction and small-scale forest clearings. This advancement has driven MTCD toward more detailed and application-specific scenarios. However, such high resolution also amplifies challenges of image registration. Even after radiometric correction (Lin et al., 2022) and geometric alignment (Li et al., 2015), pixel-level spatial misalignments may still occur due to illumination differences (e.g., solar elevation angles, shadow positions), as illustrated in Fig. 1b (Li and Wu, 2024). These discrepancies can introduce pseudo-change signals in multi-temporal comparisons, reducing detection accuracy and reliability.

Recent supervised MTCD methods based on deep learning have attempted to address the challenges of multi-temporal change detection by designing more powerful spatio-temporal architectures. For example, models incorporating 3D Swin Transformers and convolutional recurrent networks have been proposed to enhance temporal modeling (Mao et al., 2024; He et al., 2024; Wen et al., 2025), while multi-scale attention mechanisms and feature-alignment modules have been developed to mitigate spatial mismatches (Zhang et al., 2024; Li and Wu, 2024). Despite these advances, supervised approaches still rely heavily on large amounts of pixel-level annotations, which are prohibitively expensive to obtain for high-resolution and long-term multi-temporal datasets. Moreover, due to limited label diversity and strong spatio-temporal heterogeneity, their performance often degrades when applied to new geographic regions or sensors. To reduce reliance on manual labels, some unsupervised MTCD methods have been explored. For instance, Negri et al. (2024) introduced an unsupervised MTCD framework that combines stochastic distances with optimized thresholding to capture land-cover dynamics. While such methods circumvent the need for annotations, they generally lack task-specific constraints and struggle to learn discriminative representations for complex temporal patterns.

Self-supervised learning (SSL) provides a promising alternative between supervised and unsupervised paradigms, and has emerged as an increasingly popular strategy for addressing the scarcity of annotated data in remote sensing (Huang et al., 2022; Yuan et al., 2022). The key idea is to design pretext tasks that generate supervisory signals directly from unlabeled data, thereby reducing dependence on manual annotations. SSL typically consists of two stages: (1) a pretraining stage, in which a network learns general-purpose representations through pretext tasks without human-labeled data; and (2) a fine-tuning stage, in which the pretrained model is further adapted to downstream tasks such as land-cover classification or change detection using a small amount of labeled data. This two-stage paradigm enables knowledge transfer from large unlabeled datasets to label-scarce scenarios, making SSL well suited for the increasingly abundant but largely unannotated multi-source satellite archives. Among them , SSL focusing on video data processing can learn temporal features. However, existing SSL focuses on sequence rearrangement (Saha et al., 2020), mask reconstruction (Cong et al., 2022; Dumeur et al., 2024; Zhang et al., 2025) or autoregressive short-term prediction (Ren et al., 2024). These methods, while effective for downstream tasks like land cover classification, overlook gradual land feature changes in MTCD. This is because they prioritize short-term correlations, localized reconstructions, or fixed-sequence

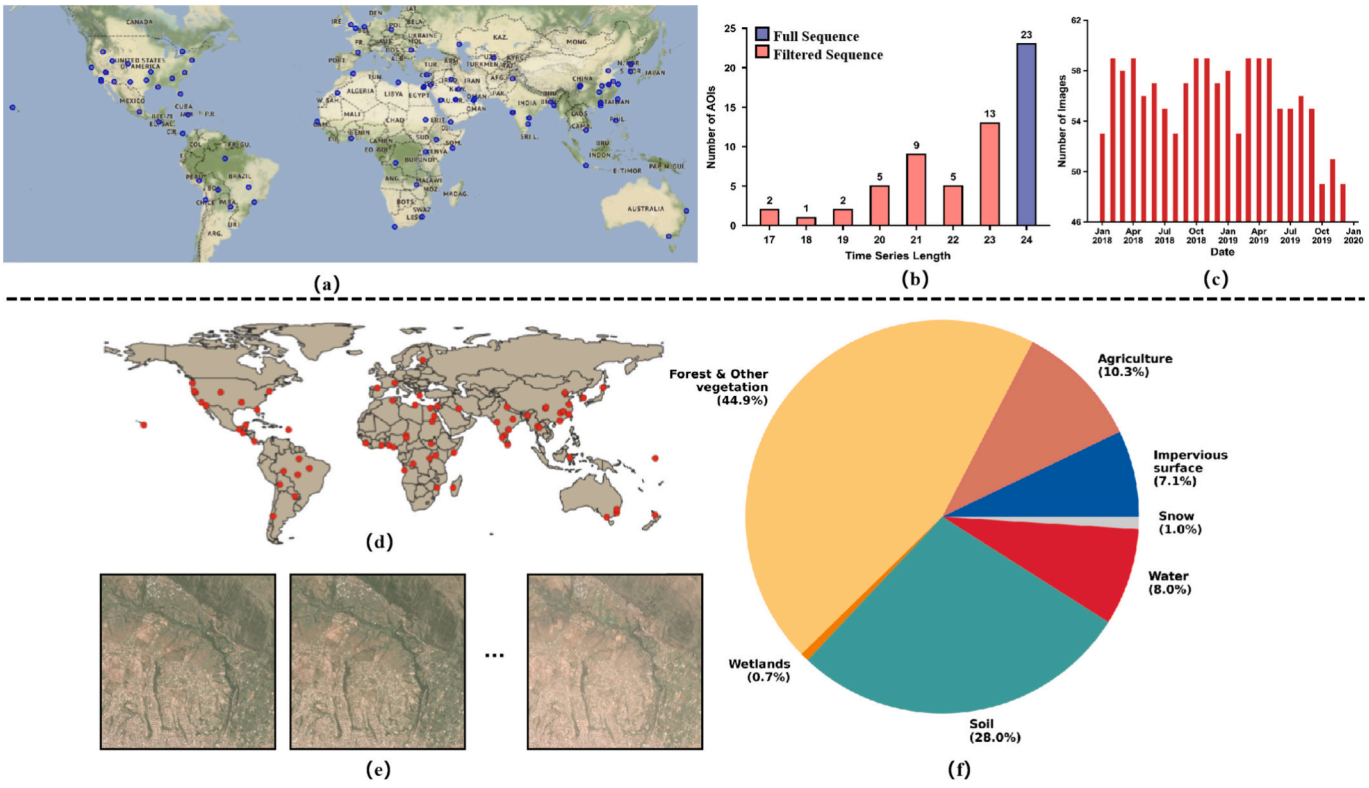


Fig. 2. Visualization of the MUDS and DynamicEarthNet dataset. (a) Locations of AOIs in the MUDS dataset, taken from Zhang et al. (2021). (b) number of AOIs corresponding to different time series lengths in the MUDS dataset. (c) monthly number of PlanetScope satellite observations in the MUDS dataset from January 2018 to December 2019. (d). Locations of all AOIs in DynamicEarthNet dataset, taken from Toker et al. (2022). (e). Three PlanetScope satellite observations from an AOI in DynamicEarthNet dataset, including 2018-01-01, 2018-02-01, and 2019-12-31. (f).Land Use Land Cover (LULC) class distribution in DynamicEarthNet dataset.

objectives that do not model the cumulative, nonlinear evolutions. As a result, they struggle to capture the subtle evolution of land cover over time.

To address the above challenges, this paper proposes a multi-task self-supervised contrastive learning network—AnyTime-CD. This approach supports flexible input of time series with arbitrary lengths and encourages the joint modeling of temporal-spectral-spatial features. Existing MTCD frameworks, while innovative, often falter in operational deployment: temporal modeling struggles with irregular sequences due to fixed-length assumptions, spectral learning is vulnerable to perturbations, and spatial alignment relies on error-prone pixel registration. AnyTime-CD overcomes these limitations through a SSL paradigm that decomposes the problem into three complementary sub-tasks: temporal dynamic modeling, spectral semantic consistency learning, and spatial feature alignment. By enabling label-free analysis of irregular, high-resolution TSI, AnyTime-CD facilitates robust, scalable deployment across globally distributed dynamic environments.

In summary, the main contributions of this study are as follows:

1) To overcome limitations in handling irregular temporal sequences, a temporal dynamic modeling task is designed to mine the temporal feature representations in satellite image time series. And, a time-dependent module is introduced to capture complex temporal dependencies and dynamic patterns within the sequence. It further decouples the mixed components in temporal signals (long-term trends, seasonal cycles, and random noise) and mitigates the impact of irregular time intervals caused by satellite revisit frequency, cloud cover, and sensor inconsistencies.
2) To mitigate spectral inconsistencies from environmental perturbations, a spectral-semantic consistency learning task with two components is proposed for remote sensing time series. Firstly, it acquires global spectral features through temporal information aggregation. Secondly, by employing two frequency-domain augmentation strategies (frequency mixing and frequency masking), it can simulate spectral perturbations while retaining basic semantic information. This enables the model to learn spectral-invariant semantic features.
3) To resolve spatial misalignment issues without relying on traditional registration, a spatial feature alignment subtask is introduced. Rather than relying on pixel-level image registration, this sub-task aligns deep semantic features across time to learn temporally invariant land cover representations. Consequently, the model exhibits improved capability in accurately localizing genuine land cover changes, while maintaining robustness against spatial distortions, cloud occlusions, and other typical disturbances in multi-temporal satellite imagery.

2. Dataset overview and preprocessing

2.1. Dataset description

To evaluate the proposed method, we conducted comprehensive experiments on two public and diverse datasets: MUDS and DynamicEarthNet (Toker et al., 2022). These datasets, which encompass extensive temporal sequences and diverse geographical extents, exhibit diversity in terms of spatial resolution, geographical distribution, and land cover characteristics, thus enabling a reliable assessment of the validity and generalization ability of our model.

The MUDS dataset consists of time series images acquired by PlanetScope satellites between 2017 and 2020 across 60 globally distributed areas of interest (AOIs), representing a diverse range of urban development patterns and intensities (as illustrated in Fig. 2). Each AOI contains a 24-month image sequence with a fixed spatial resolution of approximately 4.77 m and a uniform image size of 1024 × 1024 pixels, with three spectral bands (RGB). The reference semantic labels for each time point are categorized into two classes: building and non-building,

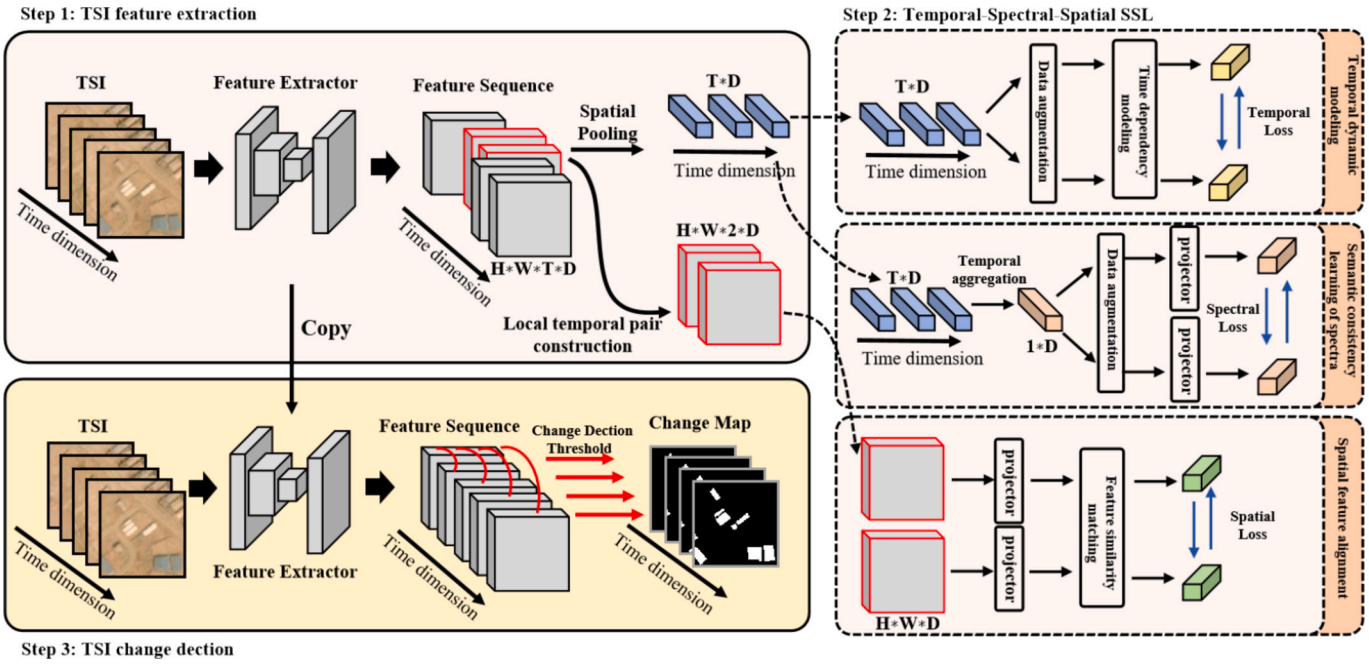


Fig. 3. An overview of the proposed AnyTime-CD. Firstly, a semantic feature extractor with shared weights independently encodes TSI at each time step, producing temporal feature representations $F = [F_1, F_2, \ldots, F_T] \in \mathbb{R}^{T\times H\times W\times D}$, where H × W represents spatial dimensions, T denotes the length of time series, and D indicates channel dimension. Next, two processing strategies are applied to the feature sequenceF: (1) Global spatial pooling generates a compact temporal representation $X = [x_1, x_2, \ldots, x_T] \in \mathbb{R}^{T\times D}$. (2) Local temporal pair construction selects image features $[Q \in \mathbb{R}^{H\times W\times D}, K \in \mathbb{R}^{H\times W\times D}]$ from adjacent time steps in the feature sequence. Subsequently, these features obtain discriminative time-spectral-spatial features through three sub-tasks: dynamic temporal modeling, spectral semantic consistency learning, and spatial feature alignment. The pre-trained feature extractor then generates a feature sequence from the TSI, and a spatial-temporal consistency threshold method, integrating spatial threshold and temporal consistency constraints, produces the change map sequence.

specifically designed for urban development monitoring. After filtering out images with excessive cloud cover or severe haze (Razzak et al., 2023), the effective time series lengths for each geographical location ranged from 17 to 24, and the processed time intervals depended on the availability of clear-sky observations.

The DynamicEarthNet dataset consists of high-resolution, cloud-free, multi-spectral time-series imagery acquired daily for two consecutive years (January 2018 to December 2019) and land-cover annotations on the first day of each month. It covers 75 AOIs across diverse global regions. Each image in DynamicEarthNet has a spatial resolution of 3 m and a fixed size of 1024 × 1024 pixels, with four spectral bands: RGB and near-infrared (NIR). The land cover labels include impervious surface, agriculture, forest & other vegetation, wetlands, soil, water, and snow. To maintain the same time interval and alignment with the land cover annotation, we selected the images of the first day of each month from DynamicEarthNet to constitute the time series images (Vincent et al., 2024).

2.2. Data processing

In the above two datasets, each image within each location has pixel-level land cover semantic annotations. These annotations were used to define the ground truth of land cover changes within a two-year observation period (i.e., comparing the semantic annotations of the initial image with each subsequent image), where a pixel was labeled as "unchanged" if it remained unchanged throughout the entire time series.

In remote sensing change detection task, a reasonable division of training-test data is crucial for evaluating model's generalization capability. Owing to the strong spatial autocorrelation in remote sensing images, traditional random sampling risks data leakage, where the model "memorizes" test set features via spatial dependencies between adjacent pixels instead of learning genuine generalization rules (Ploton et al., 2020). Although our dataset spans diverse land cover types across worldwide regions, intra-AOI similarities can still unintentionally leak spatial cues across splits if patches from the same AOI are mixed. To avoid this, we first partition the dataset into training, validation, and test subsets at a 4:1:1 ratio based on independent AOIs, ensuring geographic separation among splits. After the AOI-level division, each AOI is cropped into non-overlapping 128 × 128 image pairs using a sliding window. This two-stage strategy prevents spatial dependency leakage and ensures that the model is evaluated on entirely unseen geographic regions, enabling a more faithful assessment of cross-regional generalization.

3. Methodology

3.1. Overall framework

In this study, a multi-task self-supervised learning framework—AnyTime-CD designed for MTCD task is proposed, and the overall flow is shown in Fig. 3. The AnyTime-CD framework consists of the following three main steps:

Step 1: TSI feature extraction (see Fig. 3, top left). First, feature sequence from the input multi-temporal remote sensing images is extracted. To effectively process temporal information, each image in the sequence is independently encoded using a shared-weight encoder-decoder semantic segmentation network. This produces a feature sequence $F = [F_1, F_2, \ldots, F_T] \in \mathbb{R}^{T\times H\times W\times D}$, where H × W denotes spatial dimension, T is number of time steps, and D is feature dimension. The encoder-decoder architecture consists of two main components: (1) A Swin Transformer-based encoder (Liu et al., 2021a, 2021b), that compresses individual image into compact feature representation while capturing multi-scale contextual information. (2) The decoder, implemented via UPerNet (Xiao et al., 2018), that systematically recovers spatial details and generates output predictions with a resolution consistent with that of the input image.

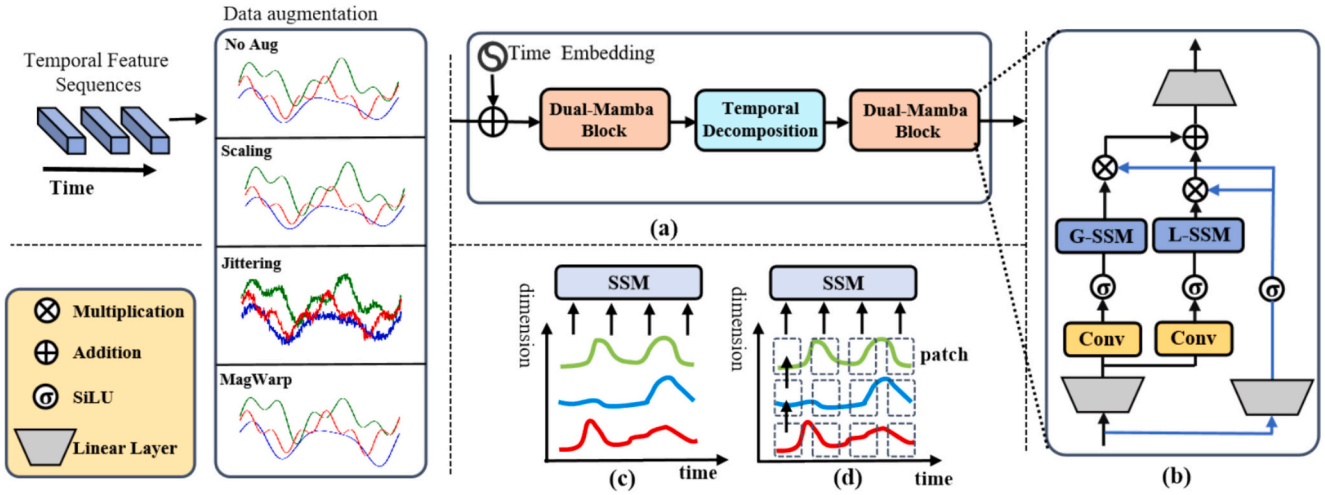


Fig. 4. Schematic diagram of Data augmentation methods and time-dependent modeling module. Data augmentation methods used in this study: Jitter, Scaling, and MagWarp. Red, green, and blue curves represent different spectral channels. Jitter: adds small random noise to each time point while preserving the waveform profile. Scaling: applies a multiplicative factor to the entire sequence, maintaining sequential order and relative variations. MagWarp: introduces nonlinear amplitude distortions while preserving the sequential order and periodicity. (a) The overall structure of the module, which integrates time encoding, Dual-Mamba, and temporal decomposition blocks. (b) Dual-Mamba structure, with the auxiliary branch indicated by blue arrows. (c) illustrates the architecture of G-SSM. (d) illustrates the architecture of L-SSM, including the temporal segmentation of the time series into non-overlapping patches of size 4. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

To capture both long-term and short-term temporal patterns, we introduce two complementary strategies: (1) Global spatial pooling: by performing spatial global pooling on feature maps across the entire time series, macro scale temporal trends (e.g., seasonal cycles and annual evolution) are modeled (Yuan and Lin, 2021). (2) Local temporal pair construction: extract neighboring timestamps from the feature sequence to guide the model to focus on short-term dynamics and fine-grained spatial details, thereby enhancing the perception ability of local spatiotemporal dynamics.

Step 2: Temporal-Spectral-Spatial SSL (see Fig. 3, right). We integrate feature learning across temporal, spectral, and spatial dimensions into a unified SSL paradigm. By designing three complementary subtasks, including temporal dynamic modeling (upper right of Fig. 3), spectral semantic consistency learning (middle right of Fig. 3) and spatial feature alignment (lower right of Fig. 3), which promote the comprehensive understanding and representation of the surface change process. These sub-tasks are optimized jointly during pre-training via the multi-task loss (Eq. 8, Section 3.5), backpropagating gradients end-to-end through the shared feature extractor to yield transferable representations.

Step 3: TSI change detection (see Fig. 3, bottom left). After completing SSL pretraining, the optimized semantic feature extractor is applied to change detection task. Specifically, using the SSL pre-trained semantic segmentation feature extractor to generate feature vectors from time series images. Then, the feature map from the first-time step serves as reference, and difference feature maps are computed for each subsequent time step by comparing bi-temporal image features. Finally, a spatial-temporal consistency threshold strategy is employed to identify spatial-temporal changes. This strategy integrates spatial threshold and temporal consistency constraints. Among them, spatial change detection generates a change map sequence by adaptively identifying the "corner" on the distribution curve of each difference feature map and calculating its maximum deviation from the straight line connecting the endpoints of the curve (Chen and Bruzzone, 2021). To enhance temporal coherence, we apply a temporal consistency constraint to the sequence of change maps. For each pixel of change map sequence, a one-dimensional label sequence is formed and processed with a sliding window of length 5. The label at the center time step is reassigned to the class with the highest frequency within the window if its occurrence exceeds half of the window length; otherwise, the original label is preserved to avoid oversmoothing. This temporal voting scheme suppresses isolated false positives induced by pseudo-changes while retaining genuine land-cover transitions that persist across multiple observations, thereby improving spatio-temporal detection stability.

It can be seen that the main part of the network is step 2. The following sections will successively introduce in detail the network structure design, training objectives, and specific implementation methods of each sub-task in step 2.

3.2. Temporal sub-task: Temporal dynamic modeling

This task is designed for a new MTCD contrastive learning strategy, aiming to mine temporal feature representations with strong discriminative ability in the satellite image time series of arbitrary lengths. The core idea of contrastive learning is to maximize the similarity between positive sample pairs (defined as a given instance and its semantically preserved augmented view) while minimizing similarity between negative pairs (defined as feature representations derived from distinct, unrelated instances within the batch), thereby learning feature representations (Chen et al., 2020). Among them, positive pairs are typically composed of a sample and its augmented view that preserves the underlying semantics, while negative pairs are drawn from different samples. Therefore, in this section, we design a data augmentation scheme and a time-dependent modeling module that meets the task requirements, and then utilize InfoNCE function (Eq. 1) to guide network training by encouraging the model to align positive pairs while separating negative pairs:

$$\mathscr{L}_{\mathrm{CL}}(\widetilde{z}_i, \widehat{z}_i) = -\log \frac{\exp(sim(\widetilde{z}_i, \widehat{z}_i)/\tau)}{\sum_{j=1}^{B}\left[\exp(sim(\widetilde{z}_i, \widehat{z}_i)/\tau) + \exp\left(sim\left(\widetilde{z}_i, \widehat{z}_j\right)/\tau\right)\right]} \tag{1}$$

$$sim(\widetilde{z}_i, \widehat{z}_i) = \frac{(\widetilde{z}_i)^{\top}(\widehat{z}_i)}{\| \widetilde{z}_i \| \| \widehat{z}_i \|} \tag{2}$$

where B is batch size, τ denotes the temperature parameter, which scales the sharpness of the softmax function (Chen et al., 2020), $[\widetilde{z}_i, \widehat{z}_i]$ and $(\widetilde{z}_i, \widehat{z}_j)$ denote output feature representation of the positive sample pair and the negative sample pair, respectively.

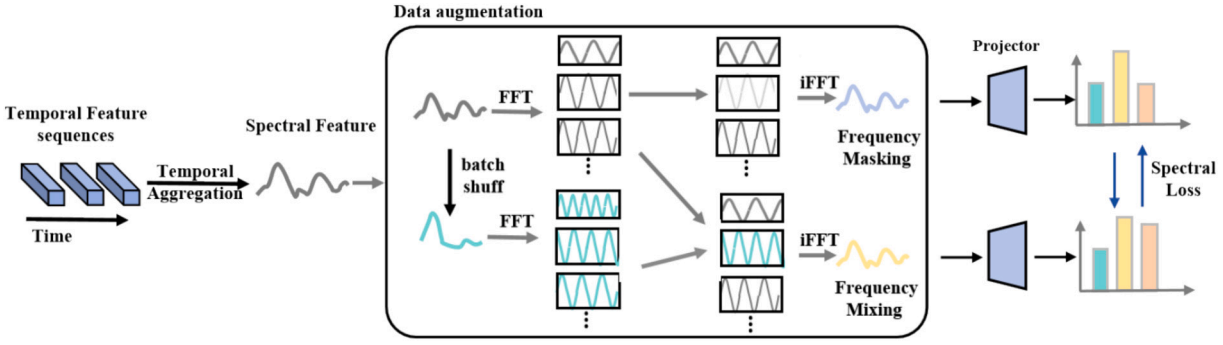


Fig. 5. Workflow of the spectral semantic consistency learning task. A temporal aggregation strategy is first applied to the global temporal feature sequences to obtain a spectral representation. Two data augmentation techniques (i.e., frequency mixing and frequency masking) are then used to generate different augmented views of the spectral features. Finally, InfoNCE is employed for optimization.

3.2.1. *Time series data augmentation*

Specifically, given the global temporal feature sequence $X = [x_1, x_2, \ldots, x_T] \in \mathbb{R}^{T\times D}$(as illustrated as the blue columns in upper right of Fig. 3), a series of randomization data augmentation techniques are applied to generate different views of the same time series (Um et al., 2017), including Jitter (adding random noise), Scaling (amplitude modification), and MagWarp (nonlinear amplitude transformation). As shown in Fig. 4, although these augmented time series differ in appearance, they fundamentally preserve the underlying temporal patterns.

3.2.2. *Time-dependent modeling incorporating seasonal cycles and long-term trends*

In MTCD, time series signals are often mixed by long-term temporal trends, seasonal cycles, and random noise. Additionally, uneven time intervals due to irregular satellite revisit periods, cloud cover, and sensor variability further complicate temporal modeling. To address these issues, global time encoding is embedded into semantic features x_t at each time step, and a Dual-Mamba block is designed to capture the time dependencies and sequential dynamics. Furthermore, frequency-domain decomposition is applied to decouple long-term trends and seasonal periodicity. The detailed process is as follows:

As shown in Fig. 4a, this module contains a temporal encoding procedure and two Dual-Mamba blocks and a temporal decomposition block.

1) Time encoding: day of year (DOY)-based positional encodings are employed for each temporal step and added to the corresponding feature vectors $X = [x_1, x_2, \ldots, x_T] \in \mathbb{R}^{T\times D}$, thereby embedding temporal positional information into the semantic representations. Specifically, given a feature sequence X, dimensional time position encoding vectors are added to each feature vector $\widetilde{x}_t = x_t + p_t$, where p_t is encoded by the year, month and day corresponding to each time step, which is defined as:

$$p_t = \left[sin\left(\mathrm{DOY}_t / r^{2i/D}\right), cos\left(\mathrm{DOY}_t / r^{2i/D}\right)\right]_{i=1}^{D/2} \tag{3}$$

where temporal scaling factor r is set to 1000 following Dumeur et al. (2024), which stabilizes the sinusoidal encoding of DOY features and ensures smooth periodic variation across time. In this manner, global temporal information is embedded into the feature sequence, which helps the network to understand irregular time intervals.

2) Dual-Mamba block: To capture time dependencies and sequential dynamics, this paper designs a variant of Mamba (Gu and Dao, 2023), i.e., Dual-Mamba block (Fig. 4b). It contains a dual-branch structure, in which the auxiliary branch enhances global-local modeling strategy in main branch via a short-cut connection, and the main branch consists of two parallel global and local sub-paths. In each sub-path, features are first passed through a 1D convolutional block and then fed into the corresponding State Space Models (SSMs). Among them, G-SSM (i.e., Global State Space Models) employs continuous-time state equations to propagate hidden states across semantic features at all time steps (Fig. 4c), enabling efficient capture of long-range temporal dependencies. And the Local-SSM (L-SSM) adopts a temporal segmentation strategy (Fig. 4d) to model local spectral-temporal interactions, encouraging the model to learn dynamic features at fine temporal granularity. Among them, inspired by the gating operation in Gated MLPs (Liu et al., 2021a, 2021b), the auxiliary branch applies an activation function to produce nonlinear modulation weights. These weights are then combined with the outputs of the G-SSM and L-SSM through element-wise multiplication, enabling dynamic gating of temporal dependencies while maintaining dimensional consistency.

3) Trend-season decomposition block: A time series decomposition based on frequency domain is adopted to extract seasonal components from each feature sequence. Specifically, the feature $\mathrm{Dual_manba}(\widetilde{x}_t) \in \mathbb{R}^{T\times D}$, derived from the time-encoded feature sequence $\widetilde{x}_t$ after processing through the first Dual-Mamba block, is transformed into the frequency domain using the Fast Fourier Transform (FFT). Although the FFT operates in the learned feature space rather than on raw pixels, the encoder preserves the temporal ordering and periodic patterns of the original time series. And Mamba has further enhanced the dynamic representation ability of sequences and improved the robustness of frequency decomposition. This ensures that low-frequency components still capture smooth, long-term temporal trends, while high-frequency components correspond to seasonal or short-term fluctuations (Rhif et al., 2022; Yang et al., 2025). This process can be formulated as:

$$\mathrm{season},\ \mathrm{trend} = \mathrm{iFFT}\left(\mathrm{FFT}(\mathrm{Dual_manba}(\widetilde{x}_t))_{[>\xi]}, \mathrm{FFT}(\mathrm{Dual_manba}(\widetilde{x}_t))_{[\leq\xi]}\right) \tag{4}$$

where $\mathrm{FFT}(\mathrm{Dual_manba}(\widetilde{x}_t))_{[>\xi]}$ represents the high-frequency component with absolute value above the threshold ξ, $\mathrm{FFT}(\mathrm{Dual_manba}(\widetilde{x}_t))_{[\leq\xi]}$ indicates the retention of the low-frequency component related to the long-term trend, and $I = T/2 + 1$. trend $\in \mathbb{R}^{I\times D}$ represents the long-term evolution of the input sequence, and season reflects periodic dynamic patterns. In this scenario, seasonal components contain substantial noise. Thus, we retain those matching the number of trend components with the highest frequency for effective denoising. Finally, the processed data are fed into the subsequent Dual-Mamba block to further learn temporal dynamics.

3.3. Spectral sub-task: spectral semantic consistency learning

Time-series features are characterized by multi-spectral observations acquired at different time steps. While cloud contamination, aerosol effects, and random sensor noise can degrade the quality of spectral observation, thereby interfering with feature representation. To mitigate these challenges, this study introduces a spectral semantic consistency learning sub-task to learn invariant representations that are robust to such perturbations, hereby preserving semantically meaningful land cover attributes (e.g., distinguishing vegetation from impervious surfaces) while effectively suppressing noise, optimized using InfoNCE (Eq. 1). It employs two frequency-domain data augmentation strategies to ensure augmented samples introduce moderate spectral variations while preserving semantic consistency, with negative samples drawn from other samples. The specific process is shown in Fig. 5.

3.3.1. Temporal aggregation

Although the global feature sequence $X = [x_1, x_2, \ldots, x_T] \in \mathbb{R}^{T\times D}$ (as illustrated as the blue columns in middle right of Fig. 3) capture spectral variation information, temporally adjacent observations are often highly correlated. This introduces significant temporal redundancy and renders them susceptible to interference from cloud cover, sensor noise, and imaging angles. To address these issues, adaptive average pooling along the temporal dimension is conducted, which can be formulated as:

$$O^k = \frac{1}{|I^k|}\sum_{t\in I^k} x_t, k = 1, 2, 3 \tag{5}$$

where I^k denotes the k-th window in adaptive segmentation. When $k = 1$, X is transformed into a stable spectral feature $O = [o_1, o_2, \ldots, o_T] \in \mathbb{R}^D$, which captures the spectral characteristics of the dominated land cover during the entire time series.

3.3.2. Frequency-domain augmentations for spectral representation

1) Frequency Mixing. Given the spectral feature O after time aggregation, the FFT is applied to convert it into the frequency domain, as expressed by $O^f = \mathrm{FFT}(O)$. Among them, low-frequency components typically represent dominant spectral features and land cover category information, while the high-frequency components reflect local spectral perturbations and detailed variations. Subsequently, sample pairs within the same training batch are randomly selected, and their high-frequency components are partially exchanged. iFFT is then applied to reconstruct new spectral features. Consequently, the reconstructed spectral features retain the original land cover's dominant characteristics (low-frequency information) while incorporating local details and perturbations from other samples in the batch (high-frequency information).
2) Frequency Masking. Similarly, the spectral feature O is transformed into the frequency domain using FFT. A subset of frequency components is then randomly masked to simulate partial information loss. Subsequently, new spectral features are reconstructed using iFFT. This strategy encourages the model to learn a robust and generalizable spectral representation by constructing a "missing information" feature representation.

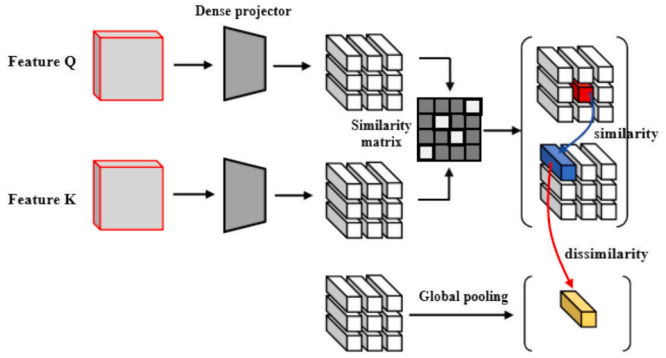


Fig. 6. Illustration of spatial feature alignment. First, bi-temporal features [$Q \in \mathbb{R}^{H\times W\times D}$,$K \in \mathbb{R}^{H\times W\times D}$] are projected into a unified representation space using a dense prediction head. Then, a similarity matrix is constructed by computing cosine similarity between all pixel vectors to establish pixel-wise correspondences. Finally, feature similarity loss and spatial consistency loss are defined based on the matching results, guiding the model to learn temporal invariant and robust representations even under spatial misalignment.

3.4. Spatial sub-task: spatial feature alignment

Multi-temporal images are often affected by factors such as orbit parameter changes, projection deformation due to terrain undulations, and preprocessing registration errors. These factors cause spatial misalignment between images acquired at different time, leading to noise in model training by aligning non-corresponding pixels, which reduces change detection accuracy. To address this, this study employ a feature matching strategy to establish pixel-level correspondences between temporally adjacent images to achieve spatial alignment, as shown in Fig. 6.

3.4.1. Feature matching

Given a pair of bi-temporal features [$Q \in \mathbb{R}^{H\times W\times D}$,$K \in \mathbb{R}^{H\times W\times D}$], we first apply a 1×1 convolution projection to perform nonlinear transformations, yielding new feature representations [$q \in \mathbb{R}^{S\times S\times d}$, $k \in \mathbb{R}^{S\times S\times d}$]. Here, d is feature dimension after projection, and $S = W = H$. Next, the projected features q and k are flattened along the spatial dimension to obtain $\widetilde{q} \in \mathbb{R}^{S2\times d}$ and $\widetilde{k}$. Pixel-wise cosine similarity is then computed between feature vectors, resulting in a similarity matrix $M \in S2 \times S2$, where M_{ij} denotes the cosine similarity between the i-th feature vector in $\widetilde{q}$ and the j-th feature vector in $\widetilde{k}$. This similarity matrix is subsequently used to identify the most relevant counterpart for each pixel in $\widetilde{q}$ via feature matching, which can be formulated as:

$$c_i = \underset{j}{\mathrm{argmax}} M_i \tag{6}$$

where j denotes the position of the matching vector of the i-th feature vector in $\widetilde{q}$ in $\widetilde{k}$, and c_i represents the similarity between the i-th feature vector in $\widetilde{q}$ and its matching vector. This matching process provides high-quality positive pairs for the subsequent comparison loss calculation, thereby enhancing the model's change detection performance under spatial misalignment conditions.

3.4.2. Spatial alignment objective

Based on the correspondence relationship of feature vectors between views extracted by feature matching, assign positive pairs [$\widetilde{q}_S$, $\widetilde{k}_S^+$], where $\widetilde{q}_S$ denotes the S-th feature vector in $\widetilde{q}$, and $\widetilde{k}_S^+$ is the positive pair with $\widetilde{q}_S$ in feature $\widetilde{k}$. The negative pair is the feature vector r^- from other

Table 1
Overview of the compared algorithms.

Types	Method	References	Characteristics
Supervised Bi-temporal Change Detection	SwinSUNet	(Zhang et al., 2022)	A Swin Transformer module extracts bi-temporal image features, with shallow and deep features fused in the decoding stage.
	ACAHNet	(Zhang et al., 2023)	A series-parallel CNN-Transformer architecture combines local perception of CNN with global modeling of Transformer.
	BIFA	(Zhang et al., 2024)	A Siamese network performs multi-scale spatial alignment of bi-temporal images by integrating channel and spatial features.
	SCanNet	(Ding et al., 2024)	A three-branch encoder-decoder network processes bi-temporal image features, using the CSWin Transformer (Dong et al., 2022) to integrate semantic and change features.
	STENet	(Pan et al., 2024)	A Siamese network extracts bi-temporal image features, with spatial change localization achieved through multi-branch modeling.
Supervised Multi-temporal Change Detection	Unet-3D	(Reichstein et al., 2019)	A U-shaped encoder-decoder architecture with 3D convolutional modules jointly models spatial and temporal features.
	SITS-SCD	(Vincent et al., 2024)	A multi-temporal U-Net-based architecture for semantic change detection in satellite image time series, featuring a resolution-specific temporal attention mechanism
	TSSCD	(He et al., 2024)	A fully convolutional architecture with stacked one-dimensional temporal convolutions models temporal dynamic features.
Self-Supervised Change Detection	SimCLR	(Chen et al., 2020), (Chen and Bruzzone, 2021)	A self-supervised contrastive learning framework, based on Zhang et al. (2021), treats bi-temporal images as positive samples and images from different spatial locations as negative samples to extract semantic representations from global features.
	SACNet	(Chen et al., 2025)	A self-supervised encoder-decoder architecture extracts semantic features from adjacent temporal images (positive samples) and different spatial locations (negative samples), refined through contrastive learning by aggregating local details and global spatial semantics.
Relevant Generative Self-Supervised Methods	MAE	(He et al., 2022)	An encoder-decoder architecture randomly masks and reconstructs single-frame images to learn spatial features from images.
	SiamMAE	(Gupta et al., 2023)	A Siamese-based encoder-decoder architecture uses historical frames to guide reconstruction of masked regions in future frames, learning temporally dependent spatial-spectral features from video imagery.

views after spatial pooling operation (see yellow vectors in Fig. 6). The spatial alignment objective for each batch can be formulated as:

$$\mathscr{L}_{spa} = -\frac{1}{S^2}\sum_{i=1}^{S^2}\left(\underbrace{sim\left(\tilde{q}_i, \tilde{k}_i^{+}\right)}_{\text{feature similarity}} + \underbrace{\left\|P_i - P_i^{+}\right\|_2}_{\text{spatial consistency}} + \underbrace{1 \Big/ log\sum_{j=1}^{S^2} exp\left(sim\left(\tilde{q}_i, r_j^{-}\right)\right)}_{\text{feature uniformity}} \right) \tag{7}$$

where P_i is spatial position of the i-th feature vector in $\tilde{q}$ in feature map q. $sim\left(q_i, k_i^{+}\right)$ is defined as the maximum value c_i in the i-th row of the similarity matrix M. $\left\|P_i - P_i^{+}\right\|_2$ represents the spatial offset between the corresponding features in the bi-temporal feature maps.

3.5. Overall loss

To effectively guide the model in learning representative spatial-temporal-spectral features from multi-temporal remote sensing data during the pre-training stage, we designed a multi-task loss function integrating multiple learning objectives, as shown in Eq. 8:

$$\mathscr{L} = \lambda_1 \mathscr{L}_{tem} + \lambda_2 \mathscr{L}_{spe} + \lambda_3 \mathscr{L}_{spa} \tag{8}$$

where $\mathscr{L}_{\text{spa}}$, $\mathscr{L}_{\text{tem}}$, and $\mathscr{L}_{\text{spe}}$ are respectively the loss functions of Spatial, Temporal, and Spectral sub-task. λ_1, λ_2, and $\lambda_3$controls the balance between $\mathscr{L}_{tem}$, $\mathscr{L}_{spe}$, and $\mathscr{L}_{spa}$.

3.6. Evaluation metrics

MTCD aims to identify dynamic changes in land cover over time evolution. The task faces two main core challenges: (1) accurately locating pixel where change occurs; (2) accurately identifying specific time point when change occurs. Therefore, to comprehensively assess the effectiveness of the proposed method, this study evaluates change detection results in terms of spatial accuracy and temporal accuracy, following the metrics suggested in Wang et al. (2022). Both dimensions use three standard metrics: producer accuracy (PA), user accuracy (UA), and F1 score, these are calculated as:

$$\text{PA} = \frac{TP}{TP + FN} \tag{9}$$

$$\text{UA} = \frac{TP}{TP + FP} \tag{10}$$

$$\text{F1} = \frac{2*\text{PA}*\text{UA}}{\text{PA} + \text{UA}} \tag{11}$$

Specifically, for evaluating spatial accuracy, spatial-PA (S-PA), S-UA, and S—F1 are used to measure model's ability to identify changing regions in spatial dimension. At this point, S-TP denotes ratio of pixels correctly predicted as changed in whole time series, S-FP refers ratio of unchanged pixels are incorrectly predicted to changed, S-TN represents ratio of unchanged pixels that are correctly predicted, and S-FN denotes ratio of changed pixels that model failed to detect.

Similarly, temporal accuracy is evaluated by temporal-PA (T-PA), T-UA, and T-F1 to assess discriminative ability of model for changing moments. Among them, T-TP denotes ratio of time points where changes are correctly predicted within temporal error margin. T-FP represents ratio of time points where incorrect predictions of change occur. T-TN indicates ratio of time points correctly identified as unchanged. T-FN represents ratio of change moments that model failed to detect. The error margin is set to ±1 frame.

Table 2
Performance comparison between AnyTime-CD and comparison methods for change detection on the DynamicEarthNet-FL dataset. The best and second-best results are highlighted in bold and underlined, respectively. All results are described in percentage (%).

Type	Method	Pretrain	Change Location			Change Moment		
			S-PA	S-UA	S-F1	T-PA	T-UA	T-F1
Sup.	SwinSUNet	√	62.47	62.59	62.53	**65.33**	55.27	**59.88**
	ACAHNet	√	<u>66.63</u>	**64.61**	<u>65.60</u>	<u>61.49</u>	<u>56.07</u>	<u>58.66</u>
	BIFA	√	62.60	60.01	61.28	61.45	54.45	57.74
	SCanNet	√	**67.85**	<u>64.23</u>	**65.99**	59.25	53.77	56.79
	STENet	√	64.01	61.43	62.69	59.68	**56.13**	57.85
	Unet-3D	√	66.01	63.97	64.97	57.59	56.05	56.81
	SITS-SCD	√	60.61	58.44	59.51	56.45	54.44	55.43
	TSSCD	–	65.97	62.78	64.34	59.45	55.61	57.47
Self-Sup. (No label)	MAE	√	60.31	58.38	59.33	58.94	53.99	56.36
	SiamMAE	√	62.28	59.76	61.00	<u>60.24</u>	53.71	<u>56.79</u>
	SimCLR	√	61.96	<u>61.92</u>	<u>61.94</u>	54.27	52.77	53.51
	SACNet	√	<u>62.95</u>	60.46	61.68	59.30	<u>53.93</u>	56.49
	AnyTime-CD	√	**71.11**	**69.39**	**70.24**	**64.34**	**59.17**	**61.65**

4. Implementation details

4.1. Model architecture and computational time

The proposed AnyTime-CD employs a Swin-T encoder and UPerNet decoder, utilizing a shared-weight encoder-decoder architecture as a semantic feature extractor for multi-temporal high-resolution remote sensing images. For each time step, the encoder first divides each image ($3 \times 128 \times 128$) into 4×4 non-overlapping patches and projects them into 96-dimensional tokens. Through four hierarchical encoding stages, the embedding dimension is progressively expanded to 768, while the spatial resolution is correspondingly reduced from 128×128 to 16×16, enabling multi-scale feature extraction. Each stage applies patch merging (2×2 average pooling) to progressively downsample spatial dimensions while expanding channels. The UPerNet decoder integrates the multi-level encoded features through lateral connections and pyramid pooling. Specifically, features from different scales are upsampled to a uniform resolution and fused using a lightweight convolutional fusion head, generating full-resolution semantic feature maps. This structure effectively preserves fine spatial details while aggregating high-level semantic information, ensuring accurate change localization at the pixel level. After decoding, the feature maps from all temporal frames are concatenated along the temporal dimension to form a unified time-series feature representation. This representation retains both spatial consistency and temporal ordering, serving as the input for subsequent temporal–spectral–spatial analysis in change detection stage.

In terms of computational efficiency, for an input sequence of $12 \times 3 \times 128 \times 128$ (12 time steps, 3 spectral bands, 128×128 spatial size), AnyTime-CD—implemented in PyTorch—processes one batch in approximately 2.3 s during pre-training on NVIDIA RTX 2080 GPU (batch size 32), following standard preprocessing steps such as normalization and augmentation.

4.2. Comparison of methods and experimental setup

Given that there is currently a lack of SSL methods specifically designed for MTCD, we selected four types of representative algorithms that have demonstrated strong performance in related domains (see Table 1).

For fair comparison, following explanations are supplemented:

1) **Extension of BTCD training strategy:** Conventional BTCD training method is unable to model long time-series information. To address this, we extend its training strategy. Specifically, a fixed reference time image (typically the earliest available image) is paired with all subsequent temporal images for network training to predict land cover changes over time. Finally, during inference, the earliest frame is used as a fixed reference and paired with each subsequent frame to generate the change map at time. This strategy maximizes the use of temporal information while ensuring temporal consistency in change detection, enabling a direct comparison with AnyTime-CD.
2) **Architectural adaptation of SSL methods:** The standard SimCLR framework pre-trains only encoder, yielding output features with a spatial resolution of 1/64 of the input, which limits its direct application in pixel-level interpretation tasks. To enable fair comparison, we adapt SimCLR to adopt the same encoder-decoder architecture as AnyTime-CD, adding a spatial pooling module on the decoding side to generate features suitable for SimCLR's contrastive learning. Similarly, SACNet uses the same encoder-decoder as AnyTime-CD. In inference, the decoder output is utilized to generate change maps, with the same TSI change detection step as in AnyTime-CD.
3) **Training strategy for the SSL methods:** We adopt a tailored pre-training data configuration for different SSL approaches: the single-temporal method (MAE) uses all available single-temporal images, the bi-temporal methods (SimCLR and SACNet) use all available image pairs, and the multi-temporal methods (SimMAE and AnyTime-CD) use image sequences sorted in chronological order. Meanwhile, all SSL models are pretrained for 100 epochs using the AdamW optimizer with a batch size of 32. The initial learning rate is set to 0.03 and scheduled with a cosine decay strategy. Finally, the optimal model selected during the pre-training process is used for the subsequent unsupervised change detection task.
4) **Scope of comparative methods:** Considering that existing MTCD methods (e.g., Unet-3D, SITS-SCD and TSSCD) require fixed input lengths, AnyTime-CD is mainly compared with BTCD and related domain self-supervised algorithms in experiments with variable-length inputs (see Section 5.2).

For the supervised training or fine-tuning process, all methods adopt a uniform training setup:batch size is 8, stochastic gradient descent (SGD) as the optimizer, 100 training epochs, and an initial learning rate of 0.0015. Among them, for the BTCD, all available bi-temporal image pairs are used for training. The MTCD method takes in time series images arranged in chronological order for the time series learning and feature extraction of the model. All models, except for TSSCD (which operates on one-dimensional time series), were initialized with ImageNet-1 K pre-trained weights to ensure a fair and consistent pre-training configuration across methods.

5. Experiments results

In this section, performance of all above methods on the fixed-length dataset are compared. Then, the change detection accuracy of AnyTime-CD and the BTCD methods on the variable-length dataset is then evaluated. Finally, the individual contribution of each sub-task of AnyTime-

Table 3
Performance comparison between AnyTime-CD and comparison methods for change detection on the MUDS-FL dataset. The best and second-best results are highlighted in bold and underlined, respectively. All results are described in percentage (%).

Type	Method	Pretrain	Change Location			Change Moment		
			S-PA	S-UA	S-F1	T-PA	T-UA	T-F1
Sup.	SwinSUNet	√	73.25	70.60	71.90	**73.74**	66.89	70.15
	ACAHNet	√	78.85	73.49	76.08	70.26	67.94	69.08
	BIFA	√	80.28	79.64	79.96	68.76	72.15	**70.41**
	SCanNet	√	78.37	75.79	77.06	69.14	71.18	70.15
	STENet	√	74.68	74.74	74.71	62.56	**72.96**	67.36
	Unet-3D	√	**82.79**	**81.14**	**81.96**	67.22	70.06	68.62
	SITS-SCD	√	78.13	75.56	76.82	63.49	59.17	61.25
	TSSCD	–	74.15	72.86	73.50	59.79	67.81	63.55
Self-Sup. (No label)	MAE	√	54.19	53.52	53.85	54.17	53.45	53.81
	SiamMAE	√	56.32	55.35	55.83	55.70	55.14	55.42
	SimCLR	√	63.69	61.64	62.65	58.41	58.62	58.51
	SACNet	√	67.31	65.41	66.34	63.35	62.36	62.85
	AnyTime-CD	√	**77.83**	**78.05**	**77.94**	**67.91**	**71.87**	**69.84**

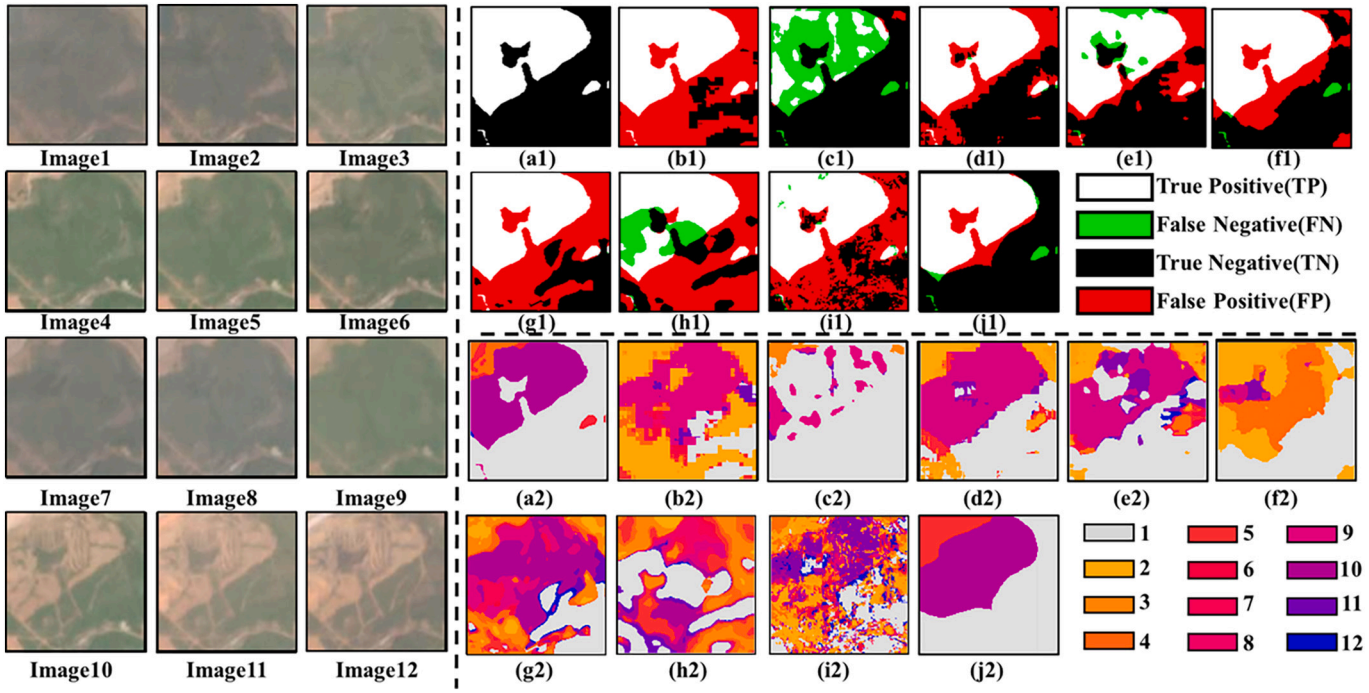


Fig. 7. Visual comparison of the change detection results obtained by different approaches on DynamicEarthNet-FL Dataset. The left is a time series image with a length of 12. The upper right presents the change location maps for (a1) Reference, (b1) SwinSUNet, (c1) ACAHNet, (d1) BIFA, (e1) SCanNet, (f1) STENet, (g1) Unet-3D, (h1) SITS-SCD. (i1) TSSCD, and (j1) Proposed AnyTime-CD. The lower right presents the final change detection timing maps for (a2) reference, (b2) SwinSUNet, (c2) ACAHNet, (d2) BIFA, (e2) SCanNet, (f2) STENet, (g2) Unet-3D, (h2) SITS-SCD, (i2) TSSCD, and (j2) Proposed AnyTime-CD, where label 1 denotes no change, and label 2–12 denotes a detected change occurred at second-12th time step, respectively.

CD is assessed via ablation experiments.

5.1. Experiments on fixed-length multi-temporal datasets

To ensure fairness, all methods are trained and tested on a time series of a uniform 12-frame image sequence length. Specifically, for the original remote sensing time series of length 24, the time frames are extracted using a fixed step size of 2. If the original sequence length is less than 24, the sampling step size is adaptively adjusted to meet the 12-frame input requirement. The experiment is conducted on the two datasets (DynamicEarthNet and MUDS), with the suffix -FL indicating the subset after extracting 12 frames following the above steps.

We evaluated BTCD, MTCD, and SSL methods on two benchmark datasets with different temporal sampling conditions, i.e., DynamicEarthNet-FL (uniform intervals) and MUDS-FL (irregular intervals). The quantitative results were summarized in Tables 2 and 3. On DynamicEarthNet-FL, AnyTime-CD achieved an S—F1 score of 70.24%, outperforming the best BTCD (S-F1: 65.99%) and MTCD (S-F1: 64.97%) methods by 4.25% and 5.27%, respectively. On MUDS-FL, AnyTime-CD attains T-F1 of 69.84% under a fully label-free setting, consistently surpassing the best MTCD baseline. Notably, compared to the top-performing SSL methods on the two datasets—SiamMAE on DynamicEarthNet-FL (S-F1: 61.00%, T-F1: 56.79%) and SACNet on MUDS-FL (S-F1: 66.34%, T-F1: 62.85%), AnyTime-CD delivered relative improvements of more than 13% in S—F1 and 8% in T-F1. Qualitative comparison also confirms the above advantages: As shown in Fig. 7, AnyTime-CD generates sharper change boundaries and significantly reduces false alarms on DynamicEarthNet-FL, while the BTCD methods (such as BIFA, SCanNet) are still disturbed by noise. Similar trends are observed in Fig. 8 for MUDS-FL, further confirming the robustness of the

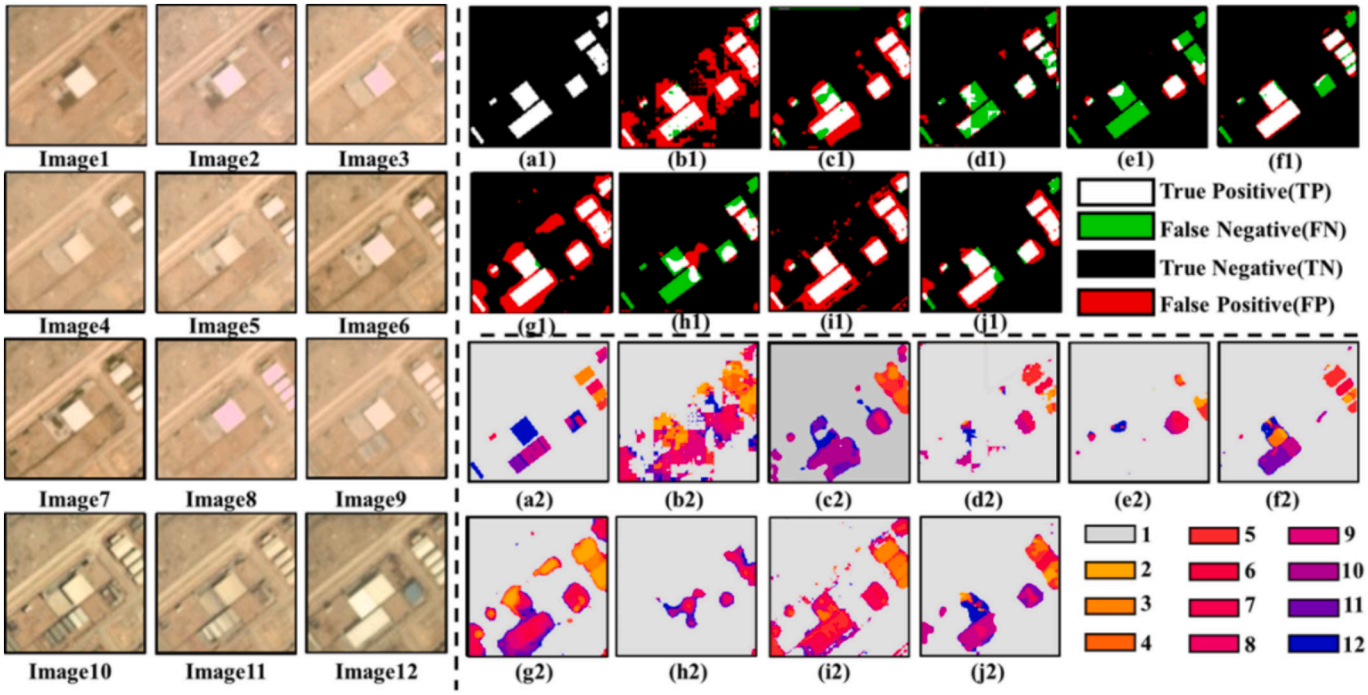


Fig. 8. Visual comparison of the change detection results obtained by different approaches on MUDS-FL Dataset. The left is a time series image with a length of 12. The upper right presents the change location maps for (a1) Reference, (b1) SwinSUNet, (c1) ACAHNet, (d1) BIFA, (e1) SCanNet, (f1) STENet, (g1) Unet-3D, (h1) SITS-SCD. (i1) TSSCD, and (j1) Proposed AnyTime-CD. The lower right presents the final change detection timing maps for (a2) reference, (b2) SwinSUNet, (c2) ACAHNet, (d2) BIFA, (e2) SCanNet, (f2) STENet, (g2) Unet-3D, (h2) SITS-SCD, (i2) TSSCD, and (j2) Proposed AnyTime-CD, where label 1 denotes no change, and label 2–12 denotes a detected change occurred at second-12th time step, respectively.

Table 4
Performance comparison between AnyTime-CD and comparison methods for change detection on the MUDS-VL2 dataset. The best and second-best results are highlighted in bold and underlined, respectively. All results are described in percentage (%).

Type	Method	Pretrain	Change Location			Change Moment		
			S-PA	S-UA	S-F1	T-PA	T-UA	T-F1
Sup.	SwinSUNet	√	73.18	69.32	71.20	66.74	65.76	67.15
	ACAHNet	√	77.06	73.01	74.98	73.12	68.45	70.71
	BIFA	√	**81.57**	**76.03**	**78.70**	70.89	72.06	**71.47**
	SCanNet	√	77.53	72.91	75.14	**74.38**	67.52	70.78
	STENet	√	74.85	72.85	73.84	67.27	**72.50**	69.78
Self-Sup. (No label)	MAE	√	54.10	53.31	53.70	61.85	57.40	59.54
	SiamMAE	√	57.87	56.57	57.22	63.07	58.40	60.65
	SimCLR	√	65.67	63.97	64.81	65.30	63.34	64.30
	SACNet	√	67.85	65.50	66.66	67.35	63.77	65.51
	AnyTime-CD	√	**80.48**	**78.85**	**79.66**	**72.36**	**72.37**	**72.36**

Table 5
Performance comparison between AnyTime-CD and comparison methods for change detection on the MUDS-VL4 dataset. The best and second-best results are highlighted in bold and underlined, respectively. All results are described in percentage (%).

Type	Method	Pretrain	Change Location			Change Moment		
			S-PA	S-UA	S-F1	T-PA	T-UA	T-F1
Sup.	SwinSUNet	√	71.72	68.71	70.18	**73.74**	66.89	**70.15**
	ACAHNet	√	69.57	**72.89**	71.19	65.25	**69.12**	67.13
	BIFA	√	70.45	69.05	69.74	69.81	68.68	69.24
	SCanNet	√	**74.10**	72.68	**73.39**	70.03	68.89	69.45
	STENet	√	63.14	65.13	64.12	63.13	66.87	64.94
Self-Sup. (No label)	MAE	√	60.70	60.36	60.53	63.07	58.40	60.65
	SiamMAE	√	58.93	58.41	58.67	62.30	62.37	62.34
	SimCLR	√	60.81	60.00	60.40	65.87	67.77	64.07
	SACNet	√	67.50	67.09	67.29	63.28	64.90	64.08
	AnyTime-CD	√	**71.98**	**71.49**	**71.73**	**71.27**	**70.39**	**70.83**

Table 6
Results of ablation experiments on the MUDS-FL dataset. The best and second best results are shown in bold and underline, respectively.

ID	Components			Change Location			Change Moment		
	$\mathscr{L}_{spa}$	$\mathscr{L}_{spe}$	$\mathscr{L}_{tem}$	S-PA	S-UA	S-F1	T-PA	T-UA	T-F1
1				59.05	60.51	59.77	55.19	59.89	57.44
2	√			70.41	69.84	70.11	60.65	67.00	63.67
3		√		75.36	<u>74.89</u>	<u>75.12</u>	63.04	<u>69.38</u>	66.06
4			√	<u>76.37</u>	72.07	74.16	<u>66.14</u>	69.15	<u>67.61</u>
5	√	√	√	**77.83**	**78.05**	**77.94**	**67.91**	**71.87**	**69.84**

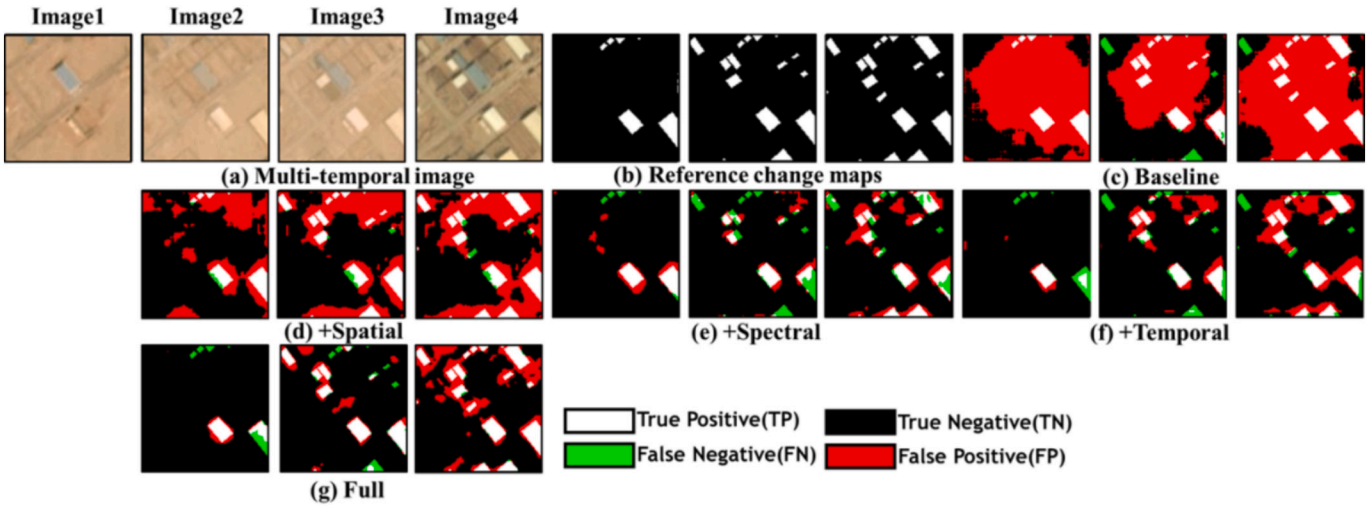


Fig. 9. Visual comparison of the confusion map obtained by ablation experiments on the MUDS-FL Dataset. (a) Image1-Image4 consists of 4 images at different times, among which image1 is the image at the initial moment, and image2–4 is the image that has changed. (b) Reference change maps, from left to right corresponding to the transitions Image1 → Image2, Image1 → Image3, Image1 → Image4. (c) Baseline (ImageNet-1 K pre-training only), (d) + Spatial (Baseline + spatial alignment), (e) + Spectral (Baseline + spectral semantic consistency), (f) + Temporal (Baseline + temporal dynamic modeling), and (g) Full (all three subtasks).

proposed method under both regular and irregular time interval conditions. This performance improvement is attributed to the adaptive spatial-temporal aggregation and implicit spatial alignment mechanisms proposed in the method, which effectively suppress redundant temporal noise, correct spatial misalignments, and focus on meaningful spectral semantic features that reflect actual land cover dynamics. Nonetheless, a modest increase in false negatives appears in mixed land-cover transition zones (see the Fig. 8(j1)), reflecting the model's conservative bias toward minimizing false positives induced by illumination or registration noise. In short, AnyTime-CD achieves state-of-the-art change detection accuracy and cleaner visual outputs across both datasets.

5.2. Experiments on variable-length multi-temporal datasets

In real-world remote sensing applications, number of available images often fluctuates due to cloud coverage, satellite revisit cycles, and preprocessing strategies. To rigorously evaluate AnyTime-CD's adaptability to variable-length sequences, we constructed two dedicated subsets from the MUDS dataset (original lengths 17–24 frames): MUDS-VL2 (step size 2, yielding 8–12 frames for dense sampling and finer temporal granularity) and MUDS-VL4 (step size 4, yielding 4–6 frames for sparser sampling to simulate coarser revisit intervals). This multi-setting design tests robustness across irregular observation densities and effective lengths, compared against BTCD and other methods limited to fixed-length inputs where applicable. Then, compared it with the BTCD method that supports input of any length.

As shown in Tables 4 and 5, the spatial and time detection performance of some supervised methods degrades as the sequence length decreases. In contrast, AnyTime-CD consistently maintains high detection accuracy and robustness, demonstrating its superior capability in modeling temporal dynamics within a sequence.

5.3. Ablation experiments

To assess the individual and synergistic contributions of the three self-supervised subtasks in AnyTime-CD, we conducted ablation experiments on the MUDS-FL subset, as summarized in Table 6. Starting from a Baseline model (pre-trained on ImageNet-1 K only), we progressively incorporated each subtask: +Spatial (Baseline + spatial feature alignment), +Spectral (Baseline + spectral semantic consistency), +Temporal (Baseline + temporal dynamic modeling), and Full (all three subtasks combined). This sequential design allows us to isolate the impact of each component on spatial and temporal change detection performance.

When only spatial alignment is involved, numerous geometric shifts related FPs are effectively reduced (see the changes from Fig. 9c to Fig. 9d), yielding substantial improvements in both spatial and temporal detection metrics, with S—F1 and T-F1 increasing by 10.34% and 6.23%, respectively. Fig. 9e shows that introducing spectral semantic consistency alone significantly reduce FPs but leaves several omissions, largely due to its role in mitigating spectral shifts from lighting and seasonal variations. Fig. 9f shows that enabling only temporal dynamic modeling can maintain consistency in detected ground object changes over time. For individual components (see Table 6), temporal dynamic modeling plays a key role. When all three sub-tasks are jointly optimized, they exhibited strong complementarity, further boosting S—F1 and T-F1 to 77.94% and 69.84%, representing cumulative improvements of 18.17% and 12.4% over the baseline. This demonstrates that the temporal-spectral-spatial collaborative SSL can simultaneously resolve spatial mismatches, spectral drifts, and temporal noises, which gives AnyTime-CD the most robust change detection capability.

Table 7
Sensitivity analysis of loss weight configurations on the MUDS-FL dataset.

	Loss Weight			Change Location			Change Moment		
	$\mathscr{L}_{tem}$	$\mathscr{L}_{spe}$	$\mathscr{L}_{spa}$	S-PA	S-UA	S-F1	T-PA	T-UA	T-F1
Baseline	1	1	1	77.83	78.05	77.94	67.91	71.87	69.84
Univariate Scaling	1	0.1	1	75.92	73.99	74.94	68.03	68.28	68.15
	1	0.5	1	77.07	75.74	76.40	70.26	67.94	69.08
	1	10	1	79.26	78.00	78.62	69.89	71.36	70.61
	1	1	0.1	76.07	74.35	75.20	68.77	68.71	68.74
	1	1	0.5	76.93	75.13	76.02	68.24	69.14	68.69
	1	1	10	78.33	77.07	77.69	69.36	70.64	69.99
Cross-Extreme Scaling	1	0.1	10	78.39	76.99	75.91	69.25	70.34	69.79
	1	10	0.1	78.18	77.51	77.84	70.03	71.40	70.70
Co-Directional Scaling	1	0.1	0.1	76.37	75.30	75.83	68.33	69.44	68.88
	1	0.5	0.5	76.94	75.69	76.31	68.46	68.73	68.59
	1	10	10	78.07	76.79	77.42	68.70	70.27	69.47

Table 8
Ablation study on temporal modeling mechanisms.

Model	Change Location			Change Moment		
	S-PA	S-UA	S-F1	T-PA	T-UA	T-F1
MLP	73.09	71.45	72.26	66.22	65.78	66.00
Transformer	74.21	72.73	73.46	66.43	66.97	66.70
Dual-Mamba	76.37	72.07	74.16	66.14	69.15	67.61

5.4. Loss balancing and training dynamics

The proposed AnyTime-CD integrates three complementary self-supervised subtasks—temporal dynamic modeling, spectral semantic consistency, and spatial feature alignment—through the multi-task loss in Eq. 8. To ensure stable optimization, it is essential to balance the contributions of these loss terms. Therefore, we conducted a systematic sensitivity study to evaluate how different weight configurations affect change detection performance. Each configuration was pretrained for 100 epochs under the same experimental setup as Section 4.2. In addition, we fixed the temporal loss weight (i.e., weight of $\mathscr{L}_{tem}$= 1) as the anchor, while systematically varying the spectral and spatial loss weights across univariate, cross-extreme, and co-directional scaling configurations. This design enabling direct assessment of how spectral and spatial loss affect change detection performance.

Table 7 summarizes the performance metrics across 12 configurations. The baseline setting achieves strong performance, confirming effective synergy among the three subtasks without dependency on heavy tuning. In univariate scaling, increasing the spectral loss weight (i.e., weight of $\mathscr{L}_{spe}$= 10) produces the highest spatial detection performance (e.g., S-F1: 78.62%), whereas decreasing it causes the most pronounced performance drop, indicating that spectral semantic consistency is essential for stable representation learning and reliable change detection. And, the spatial loss similarly cannot be reduced excessively (i.e., weight of $\mathscr{L}_{spa}$= 0.1), as a weakened spatial constraint increases local misregistration-induced noise (i.e., S-F1: 77.94% to 75.20%). Notably, in cross-extreme settings, even when the spatial loss weight is reduced, performance remains stable if spectral consistency is strengthened, further emphasizing the primary role of spectral semantics consistency in forming robust spatio-temporal representations. For Co-directional scaling, simultaneous reduction of spectral and spatial constraints degrades both S—F1 and T-F1. In summary, with the temporal loss weight fixed at 1, stable and even improved optimization is achieved when the spectral loss weight is set between 1 and 10 and the spatial loss weight around 1.

5.5. Effects of dual-mamba on temporal modeling

To further validate the design choice of using Dual-Mamba for temporal dependency modeling in AnyTime-CD, we conduct an ablation study comparing three temporal modeling mechanisms: MLP, Transformer, and the proposed Dual-Mamba. This analysis is performed on the temporal-dynamics sub-task only (i.e., the "+Temporal" in Section

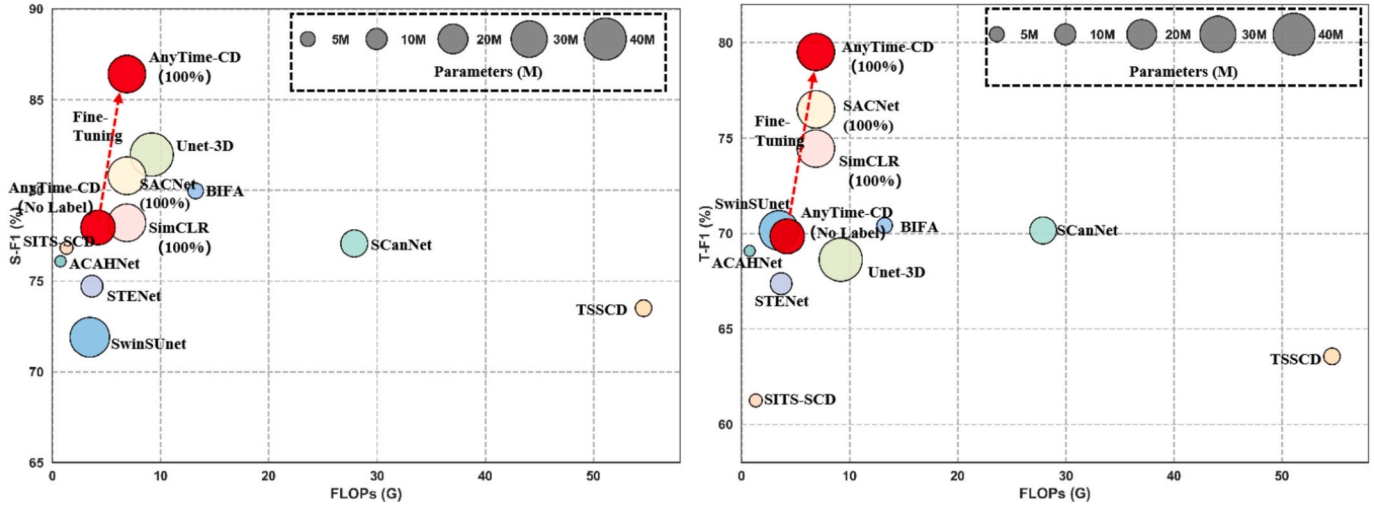


Fig. 10. Performance comparison of different change detection models on the MUDS-FL dataset, and the relationship among evaluation metrics (S—F1 and T-F1), computational efficiency (FLOPs), and model complexity (number of parameters). All models are evaluated using bi-temporal image pairs with a size of 3 × 128 × 128 (3 spectral bands, 128 × 128 spatial resolution). Different colors represent different Change detection methods.

Table 9
Results AnyTime-CD using limited label data on the MUDS-FL dataset.

Method	Ratio	Change Location			Change Moment		
		S-PA	S-UA	S-F1	T-PA	T-UA	T-F1
AnyTime-CD	0	77.83	78.05	77.94	67.91	71.87	69.84
	20	82.33	78.13	80.17	77.32	69.23	73.05
	50	87.47	84.83	86.14	80.75	77.08	78.88
	100	88.07	84.78	86.40	80.70	78.34	79.50
Best Supervised BTCD Network	100	80.28	79.64	79.96	68.76	72.15	70.41

Table 10
Results of AnyTime-CD using different pre-trained strategies.

Strategy	Change Location			Change Moment		
	S-PA	S-UA	S-F1	T-PA	T-UA	T-F1
1	77.83	78.05	77.94	67.91	71.87	69.84
2	79.60	77.69	78.33	71.17	68.88	69.98
3	72.83	72.14	72.48	66.87	66.83	66.85
4	79.40	78.38	78.88	70.33	72.27	71.29

5.3), allowing us to isolate the contribution of the temporal module without interference from the other components.

Table 8 summarizes the multi-temporal change detection results for the three mechanisms. It can be seen that MLP, which captures only shallow temporal information, yields the lowest performance. Transformer improves performance by modeling inter-temporal correlations. In contrast, Dual-Mamba achieves the best overall results, benefiting from its linear-time recurrence formulation and ability to capture both global and local temporal dependencies.

6. Discussion

6.1. Adaptability evaluation of fine-tuning under different annotation ratios

SSL algorithms can further enhance model performance with aid of annotations. To evaluate the adaptability of AnyTime-CD under different annotation ratios, we fine-tuned the pretrained model on the MUDS-FL training set using 20%, 50%, and 100% of the labeled samples, respectively. A lightweight classification head (two convolutional layers followed by activation functions) was appended to the network for fine-tuning. All training settings were kept consistent with those used in the fully supervised methods introduced in Section 5.1.

Fig. 10 shows the relationship between the performance of each method with the number of parameters and FLOPs. After fine-tuning, AnyTime-CD significantly outperformed other supervised methods on both S—F1 and T-F1. Table 9 summarizes the performance of MUDS-FL with different labeling ratios. As the labeling ratio increased from 20% to 50%, the model performance improved significantly. However, the gain tends to level off from 50% to 100% of the sample ratio, indicating that a moderate amount of labeling is sufficient for the model to learn the core features. Notably, the updated fine-tuning results for SimCLR and SACNet—obtained under the same experimental setup as AnyTime-CD—reveal that these self-supervised methods surpass supervised baselines at 100% labeling. However, both still lag behind AnyTime-CD, underscoring the superior efficiency of our multi-task design. In particular, with full labeled data, AnyTime-CD outperforms the strongest supervised baseline by 4.44% in S—F1 and 9.09% in T-F1, showcasing enhanced spatial and temporal detection capabilities. Overall, AnyTime-CD demonstrates excellent scalability and robustness across unsupervised and supervised regimes.

6.2. Effect of Pretraining data size on model performance

In SSL frameworks, the amount of pre-training data is considered as a key factor in improving SSL performance. For this reason, we designed three pre-training strategies on MUDS-FL:

1. Train+Val: replicates the setup of Section 5.1 using only the training and validation sets.
2. Full Local: adds the test set of MUDS-FL to simulate a scenario where all local images are accessible.
3. Domain-Shift Pretraining: Pre-trains exclusively on the cross-domain DynamicEarthNet-FL training and validation sets to evaluate the

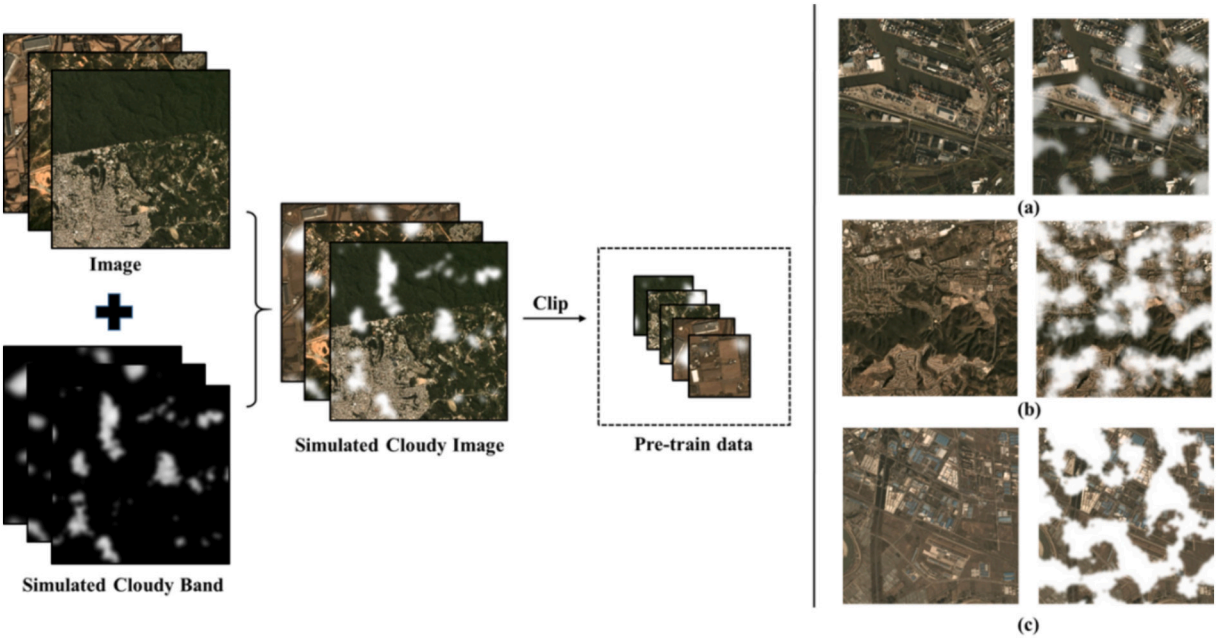


Fig. 11. Workflow of synthetic cloud image generation. A synthetic cloud image is created by overlaying a simulated cloud band onto a real remote sensing image. The resulting image is then divided into non-overlapping 128 × 128 patches for self-supervised pre-training. (a) thin cloud, (b) Semi-transparent cloud, and (c) thick cloud.

Table 11
Effects of cloud coverage on AnyTime-CD performance on the MUDS-FL dataset. "Baseline" denotes SSL using only spatial feature alignment task, "cloud mask" indicates the use of cloud mask data, and thin, semi-transparent, thick cloud refer to the simulated cloud in addition to the original time-series images.

Cloud	Cloud Type	S-PA	S-UA	S-F1
10%	Baseline	70.41	69.84	70.11
	Cloud mask	69.52 (−0.89)	69.06 (−0.78)	69.29 (−0.82)
	Thin cloud	70.26 (−0.15)	67.83 (−2.01)	69.02 (−1.09)
+10%	Semi-transparent cloud	69.08 (−1.33)	68.41 (−1.43)	68.75 (−1.36)
	Thick cloud	70.32 (−0.09)	67.68 (−2.16)	68.98 (−1.13)
+15%	Cloud mask	69.92 (−0.49)	67.68 (−2.16)	68.78 (−1.33)
	Thin cloud	70.12 (−0.29)	68.83 (−1.01)	69.47 (−0.64)
	Semi-transparent cloud	70.28 (−0.13)	69.38 (−0.46)	69.83 (−0.28)
	Thick cloud	70.91 (+0.50)	70.10 (+0.26)	70.50 (+0.39)
+20%	Cloud mask	68.95 (−1.46)	68.27 (−1.57)	68.61 (−1.50)
	Thin cloud	70.90 (+0.49)	68.37 (−1.47)	69.61 (−0.50)
	Semi-transparent cloud	71.47 (+1.06)	69.03 (−0.81)	70.23 (+0.12)
	Thick cloud	71.25 (+0.84)	69.61 (−0.23)	70.42 (+0.31)
+25%	Cloud mask	68.92 (−1.49)	66.59 (−3.25)	67.73 (−2.38)
	Thin cloud	71.61 (+1.20)	70.10 (+0.26)	70.84 (+0.73)
	Semi-transparent cloud	71.81 (+1.40)	70.68 (+0.84)	71.24 (+1.13)
	Thick cloud	72.36 (+1.95)	71.14 (+1.30)	71.74 (+1.63)
+30%	Cloud mask	68.76 (−1.65)	67.88 (−1.96)	68.31 (−1.80)
	Thin cloud	70.26 (−0.15)	67.83 (−2.01)	69.02 (−1.09)
	Semi-transparent cloud	69.84 (−0.57)	67.61 (−2.23)	68.70 (−1.41)
	Thick cloud	67.73 (−2.68)	65.30 (−4.54)	66.49 (−3.62)
	Cloud mask	68.05 (−2.36)	66.84 (−3.00)	67.44 (−2.67)

model's transferability from a heterogeneous source domain in the absence of any MUDS data.

4. Cross-Dataset: incorporates DynamicEarthNet-FL alongside MUDS-FL to evaluate cross-dataset generalization ability.

As shown in Table 10, the performance of AnyTime-CD improved steadily as the pre-training data increases. Even with only the Train+Val setting, the model achieved competitive results, indicating strong generalization ability under limited data conditions. The Full Local setting further enhances performance. Domain-Shift Pretraining yields the lowest performance, reflecting inherent cross-domain knowledge transfer challenges. Nevertheless, the framework maintains adequate detection performance, validating its domain transferability. In the Cross-Dataset scenario, rich and diverse time series remote sensing data significantly strengthen the model's capture of the spatiotemporal evolution patterns, leading to the best overall performance. These findings indicate that incorporating as diverse and time-rich remote sensing images as possible during pre-training can not only enhance the change feature representation ability but also improve the robustness and transfer performance of the model across complex environments and different time scales.

6.3. Effect of cloud interference on self-supervised change detection

Atmospheric interference, especially the cloud coverage phenomenon, is always one of the important factors affecting the image quality and the accuracy of subsequent analysis. Although many cloud-contaminated scenes are usually filtered out during image preprocessing, residual cloud interference may persist in certain images (Rußwurm and Körner, 2018), thereby interfering with the change detection results. Therefore, this section will explore the feasibility and effectiveness of incorporating cloud coverage information in the SSL pre-training stage of AnyTime-CD.

In MTCD, irregular cloud layers often degrade model performance. To explore this problem in SSL scenarios, we merged cloud masks by computing the union of cloud-covered regions (derived from image metadata) across adjacent timestamps, and then calculated the spatial loss (i.e., spatial feature alignment task, which emphasizes local spatial consistency and is sensitive to cloud-induced noise) on cloud-free areas, thereby reduce the impact of unreliable feature comparisons. To further investigate the model's robustness under different degrees of cloud coverage interference, referring to the peration of Czerkawski et al. (2023), we introduced additional 10%, 15%, 20%, 25%, and 30% of simulated cloud cover on top of the original dataset (which contains ~10% cloud cover), as illustrated in Fig. 11. To be specific, we first generated cloud bands with cloud coverage ranging from 10% to 50%, mimicking common cloud features observed in natural remote sensing scenes. In addition, the simulated cloud cover exhibited a transparency gradient from the edges toward the center, thereby simulating the physical structure of real cloud profiles. Next, the simulated cloud band are superimposed in a randomized manner into each band of the original remote sensing image to ensure that the cloud distribution is spatially non-uniform while controlling the proportion of cloud-covered pixels in the dataset. To comprehensively assess the impact of different cloud disturbances types on SSL strategies, we evaluated the model's performance under three representative cloud conditions (Zhu and Woodcock, 2012; Zhu et al., 2015; Czerkawski et al., 2023): (1) thin clouds, with alpha values ranging from 0.2 to 0.4, representing slightly occluding clouds; (2) medium-opacity clouds, with alpha values between 0.4 and

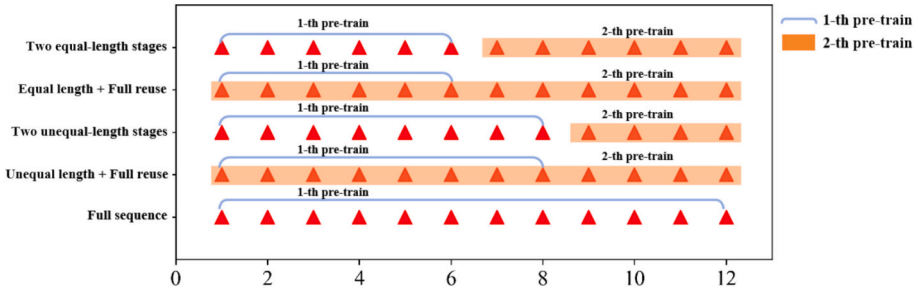


Fig. 12. Schematic illustration of the five pre-training strategies. The first pre-training stage includes 100 epochs, followed by a second stage with 10 epochs. All other training parameters, as well as the test dataset and sequence length, are consistent with those described in Section 5.1.

Table 12
Experimental results of AnyTime-CD under different training strategies on the MUDS-FL dataset.

ID	AnyTime-CD						Unet-3D					
	Change Location			Change Moment			Change Location			Change Moment		
	S-PA	S-UA	S-F1	T-PA	T-UA	T-F1	S-PA	S-UA	S-F1	T-PA	T-UA	T-F1
1	76.26	76.73	76.49	68.25	70.69	69.45	84.72	80.77	82.70	64.8	68.17	66.44
2	78.99	77.52	78.25	70.08	70.51	70.29	–	–	–	–	–	–
3	76.15	74.83	75.49	65.96	69.50	67.68	–	–	–	–	–	–
4	76.31	76.28	76.29	68.88	69.03	68.95	–	–	–	–	–	–
5	77.83	78.05	77.94	67.91	71.87	69.84	82.79	81.14	81.96	67.22	70.06	68.62

0.8, simulating medium occlusion; (3) thick clouds, with alpha set to 1.0, representing complete occlusion. Here, alpha denotes the transparency level of clouds and ranges from 0 (fully transparent) to 1 (fully opaque).

As shown in Table 11, the proposed spatial alignment task maintained stable performance under medium cloud coverage (≤30%). This indicates that the time series images are low-rank, and the structural information of the time series images can still be maintained under a certain percentage of cloud cover. It can be seen that the cloud masking approach, while effective in supervised learning for mitigating cloud interference, exhibited reduced performance in SSL under low cloud cover conditions. Furthermore, Cloud masking approach provides slight mitigation of model degradation only when an additional 30% thick cloud is involved. This suggests that excessively high cloud amounts disrupt the spatial-temporal structure of the time series images, and masking can only partially alleviate this degradation. In contrast, experiments incorporating 15% ~ 25% simulated cloud cover demonstrate improved detection accuracy. This suggests that the SSL framework can ground the cloud-induced spectral perturbations as a form of data augmentation, facilitating the learning of spatial invariance and enhancing generalization under real-world noise conditions.

6.4. Analysis of model update mechanisms for incremental remote sensing data

In practical remote sensing monitoring applications, as new images are continuously acquired, incremental model updates are required. Existing strategies typically fall into two categories: 1) Full-sequence retraining (e.g., Xu et al., 2020): retraining the model from scratch using the complete time series, achieving optimal performance but at high computational cost, which makes it unsuitable for large-scale incremental updates. 2) Fixed-window fine-tuning (Zhu et al., 2017): updating the existing model using a fixed-length sequence segment, offering improved efficiency. However, it lacks flexibility and may degrade detection performance due to the difficulty of matching the new data length to the window. To evaluate the flexibility of AnyTime-CD, we designed five training strategies (Fig. 12) based on the MUDS-FL dataset, to simulate different data availability scenarios:

Strategy 1: "Two equal-length stages": divide the complete sequence into two segments equally, train the first segment, and then incrementally update the latter segment;

Strategy 2 "Equal-length + Full-reuse": the first stage is the same as Strategy 1, but in the second stage, the model is fine-tuned using the full sequence;

Strategy 3: "Two unequal-length stages": the sequence is split into two segments of unequal length to simulate asymmetrical data accumulation;

Strategy 4: "Unequal-length + Full-reuse": based on strategy 3, reuse the full sequence in the second stage;

Strategy 5: "Full-sequence": train directly with the complete sequence as a reference.

Table 12 compares the spatial (S—F1) and temporal (T-F1) detection performance of AnyTime-CD and Unet-3D under the five training strategies. It can be seen that Unet-3D can only run in Strategy 1 and Strategy 5, and its performance in detecting change time is always lower than the baseline of Strategy 5. In contrast, AnyTime-CD demonstrates robust performance across all strategies. Notably, its T-F1 of Strategy 1 (69.45%) closely matches that of full-sequence training (Strategy 5, 69.84%), indicating that the model effectively captures temporal dynamics even with segmented data. Strategy 2 further enhances performance by combining local training with global fine-tuning, achieving the highest S—F1 (78.25%) and T-F1 (70.29%) with only a few extra epochs. Strategy 4 also shows a small improvement in temporal detection compared to Strategy 3 (68.95% vs. 67.68%), suggesting that the reuse of more time series data in the second stage helps to capture a more comprehensive dynamic. Overall, AnyTime-CD offers an efficient and effective solution for incremental updates in large-scale MTCD, without the need for full retraining.

7. Conclusion

SSL has demonstrated great potential across various remote sensing tasks. However, its application to MTCD remains in the early stages of exploration. Therefore, this paper proposed an AnyTime-CD framework, incorporates a multi-task learning strategy that jointly models temporal dynamics, spectral semantic consistency, and spatial feature alignment to capture comprehensive temporal -spectral-spatial representations. Experimental results on two widely-used large scale MTCD datasets, MUDS and DynamicEarthNet, showed that AnyTime-CD outperforms other supervised baselines across multiple evaluation metrics without relying on any labeled data. Importantly, the proposed AnyTime-CD framework provides several advantages that extend beyond algorithmic performance and have implications for multi-temporal environmental monitoring. First, its self-supervised design significantly reduces the need for dense pixel-level annotations of change locations and timing, which are often prohibitively expensive to obtain for long and irregular satellite time series. This enables large-scale change analysis in regions where annotated datasets are limited or unavailable. Second, the ability to handle arbitrary-length and irregularly sampled sequences enables the model to operate effectively under realistic, often irregular remote-sensing observation conditions. Third, its incremental processing capabilities enable the seamless integration of new observations, thereby supporting continuous updates in multi-temporal analyses.

Building on these advantages, AnyTime-CD suitable for: (i) large-scale or data-sparse regions lacking annotated change labels; (ii) monitoring regions with sparse or uneven temporal coverage; and (iii) incremental multi-temporal monitoring scenarios requiring continuous integration of new observations (e.g., vegetation succession, shoreline migration, and wetland evolution). Collectively, these application scenarios highlight the broader scientific utility of the proposed framework and its strong potential to support long-term time-series analysis in remote sensing of the environment.

CRediT authorship contribution statement

Yang Qu: Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation. **JiaYi Li:** Writing – review &

editing, Visualization, Resources, Methodology, Funding acquisition. **Xin Huang:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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